7. Technical Debt Analysis

Objective: Establish the prerequisites for an Agentic Harness & Marketplace initiative — code repository health, third-party tool usage, AI tool usage, database usage, and development environment readiness.

Date: July 15, 2026 | **Scope:** `shende-shweta/FSDKC@main` — Laravel 12 (PHP 8.3), React 19.2 + TypeScript + Vite 6, Express dev-api (Node 22), MariaDB 11, MongoDB 7, Docker Compose, GitHub Actions CI

Executive Summary

EXECUTIVE SUMMARY

Analysis covered **87 repository files** from `shende-shweta/FSDKC@main` via GitHub REST API (recursive tree + raw content fetch), including `.github/workflows/ci.yml`, root/frontend/dev-api lock files, `backend/composer.json`, Docker Compose stack, manual SQL schema (`docker/mariadb/init.sql`), and five `AGENTS.md` guides. The Klearcom monorepo has a working Docker path and partial CI (PHPUnit + frontend build), but **agentic-harness readiness is High Risk** today. The three most severe gaps are: **(1) missing `backend/composer.lock` and incomplete CI** — no lint/static-analysis gate, dev-api excluded from CI, and no branch-protection signals (`.github/CODEOWNERS`, PR template); **(2) manual flat SQL schema with no Laravel migrations** — all five MariaDB tables live in one init script with non-idempotent `INSERT` seeds and 80% cross-domain table sharing; **(3) declared-but-unenforced quality tooling** — PHPStan in `backend/composer.json:14-16` has no config file and no CI step, while ESLint/Prettier/pre-commit are absent entirely. AI-assisted development is partially bootstrapped via root and module `AGENTS.md` files plus `docs/CODEBASE_AUDIT_ISSUES.md`, but the parallel Laravel + dev-api runtimes and ~0% effective test coverage mean agents cannot safely verify refactors before merge.

Yes	1	10 / 11	Partial
TOP-LEVEL .GITIGNORE PRESENT	CI/CD WORKFLOWS FOUND	THIRD-PARTY PACKAGES DECLARED / WIRED	.ENV.EXAMPLE PRESENT

OVERALL CODEBASE RATING — TECHNICAL DEBT & AGENTIC READINESS

High Risk

Driven by High-Risk Code Repository Health (D1), Database Usage (D4), Observability Baseline (D6), and CI/Test Gate for Agent Output (D7).

Readiness Benchmark Ratings

#	Dimension	GOOD	MODERATE	HIGH RISK	Measured	Rating

D1	Code Repository Health	all checks pass	1–2 gaps	3+ gaps / no CI	`.gitignore` present but incomplete; CI in `.github/workflows/ci.yml:1-58` runs PHPUnit + frontend build only; no `backend/composer.lock` (404 on fetch); no `.github/CODEOWNERS` or PR template	HIGH R
D2	Third-Party Tool Usage	mostly wired & current	some unused/unwired	many unused/unmaintained	10/11 security/infra packages wired; `phpstan/phpstan` declared in `backend/composer.json:15` but no `phpstan.neon` and no CI invocation	MODE
D3	AI Tool / Agentic Readiness	ready	partial	not ready	5 `AGENTS.md` files + `docs/CODEBASE_AUDIT_ISSUES.md` enumerate work; `.kiro/` minimal; dual Laravel/dev-api runtime and weak CI block safe agent refactors	MODE
D4	Database Usage	sound	some gaps	no constraints / shared flat schema	Manual `docker/mariadb/init.sql:1-89` only — no migrations; FK on 2/4 domain tables; non-idempotent SQL seeds; flat shared schema across Discovery + Connect	HIGH R
D5	Development Environment	reproducible	partial	manual / fragile	`backend/.env.example` + `dev-api/.env.example` present; no `frontend/.env.example`; Docker Compose + Dockerfiles present; TypeScript check via build only — no ESLint/Prettier/pre-commit/PHPStan in CI	MODE
D6	Observability / Logging Baseline (additional)	structured logging + health endpoints	partial console logging	no logging baseline	Zero `Log::`, Monolog, Winston, Pino, or Sentry usage in 50 source files; dev-api uses `console.log` only (`dev-api/src/server.js:273-275`)	HIGH R
D7	CI / Test Gate for Agent Output (additional)	>80% critical-path coverage + lint in CI	partial gates	no effective gate	CI excludes dev-api; 2 PHPUnit files with 0% effective `App\` coverage; no frontend tests; no lint/SAST/audit steps in `.github/workflows/ci.yml`	HIGH R

D6 and D7 are additional readiness dimensions identified during this scan beyond the standard D1–D5 set.

7.1 Code Repository

Check	Files Inspected	Finding	Consequence	Next Step
`.gitignore` coverage	`.gitignore:1-9`	Covers `node_modules/`, `backend/vendor/`, `backend/.env`, `dev-api/.env`, `frontend/dist/`,	Minor risk of committing OS junk or local SQLite artifacts;	Add `.DS_Store`, `*` `.vscode/` to `.git`

		`*.log`. Missing: `.DS_Store`, `*.sqlite`, explicit `backend/storage/` beyond logs, IDE dotfiles.	not blocking but below typical monorepo hygiene.	
CI/CD presence	`.github/workflows/ci.yml:1-58`	One workflow with `backend` job (PHP 8.3, MariaDB service, `composer install`, `phpunit`) and `frontend` job (`npm ci || npm install`, `npm run build`). Triggers on push to `main` / `master` and all PRs.	Changes merge with PHPUnit + TypeScript build verification for Laravel and frontend, but dev-api (18 routes) and all lint/static-analysis tooling are ungated.	Add `dev-api` job wi PHPStan + ESLint ste `.github/workflows`
Branch protection signals	Repo tree (87 files)	No `.github/CODEOWNERS`, no `pull_request_template.md`, no `required-checks` config visible locally.	Unreviewed or untested changes can land on `main` without mandatory review or status checks beyond default GitHub settings.	Add `.github/CODEO` `.github/pull_requ` enable required status `backend` and `fron` branch settings.
Lock files committed	Tree scan + HTTP 404 on `backend/composer.lock`	`package-lock.json` committed at root, `frontend/package-lock.json`, and `dev-api/package-lock.json`. **`backend/composer.lock` absent** (HTTP 404).	PHP dependency versions float on every `composer install`; CI and local dev can resolve different Laravel/MongoDB driver patch versions, causing "works on my machine" failures.	Run `composer upda` `backend/` and comr `backend/composer.` that lock file is present

7.2 Third-Party Tools Usage

Package	Declared	Actually Wired?	Debt
`laravel/framework` ^12.0	Y (`backend/composer.json:8`)	Y — 17 PHP files import `Illuminate\` namespaces	Current major; core runtime
`mongodb/mongodb` ^2.0	Y (`backend/composer.json:9`)	Y — `backend/app/Services/MongoService.php`	Wired for transcripts/diagnostic
`mongodb` ^6.12.0 (npm)	Y (`dev-api/package.json:17`)	Y — `dev-api/src/mongo.js:1-167`	Atlas + in-memory fallback
`mongodb-memory-server` ^10.1.4	Y (`dev-api/package.json:18`)	Y — `dev-api/src/mongo.js:34-41` when `MONGODB_URI` unset	Dev-only; correctly scoped
`express` ^4.21.2	Y (`dev-api/package.json:16`)	Y — `dev-api/src/server.js` (18 route handlers)	Parallel runtime duplicates Laravel

`cors` ^2.8.5	Y (`dev-api/package.json:14`)	Y — `dev-api/src/server.js` default `cors()` ; `backend/config/cors.php:6` wildcard	Security debt: allows all origins
`dotenv` ^16.4.7	Y (`dev-api/package.json:15`)	Y — `dev-api/package.json:9-11` `--env-file=.env` scripts	Correctly wired
`phpunit/phpunit` ^11.0	Y (`backend/composer.json:14` , dev)	Y — `backend/tests/Unit/*.php` ; CI step `.github/workflows/ci.yml:47-48`	Tests exist but do not import application code
`phpstan/phpstan` ^2.0	Y (`backend/composer.json:15` , dev)	**N** — no `phpstan.neon` , no CI step, zero `phpstan` references in repo	Declared but completely unwired – dead dependency signal
`@tanstack/react-query` ^5.62.0	Y (`frontend/package.json:14`)	Y — 5 frontend files (`frontend/src/main.tsx:3-8` , pages)	Properly adopted
`zustand` ^5.0.2	Y (`frontend/package.json:18`)	Y — `frontend/src/store/uiStore.ts` , Connect/Discovery pages	Global UI selection state
`concurrently` ^9.1.2	Y (`package.json:10` , root dev)	Y — `package.json:6` `npm run dev` script	Dev orchestration only; acceptable

7.3 AI Tool Usage & Agentic Readiness

Signal	Files Inspected	Finding
AI agent guides	`AGENTS.md:1-28` , `backend/app/Modules/Discovery/AGENTS.md:1-28` , `backend/app/Modules/Connect/AGENTS.md` , `frontend/src/modules/Discovery/AGENTS.md` , `frontend/src/modules/Connect/AGENTS.md`	**Mature partial:** Five module-scoped guides document endpoints, data stores, and conventions (e.g., "No `extract()` in PHP"). Root `AGENTS.md:24-27` explicitly bans `extract()` and class components — but code still violates both (`LegacyReportController.php:22` , `LegacyMonitorPoller.jsx`).
IDE / AI tooling config	Tree scan	`.kiro/settings/mcp-bundles/.gitignore:1` only — no `.cursor/` , `CLAUDE.md` , `.github/copilot*` , or codegen scripts. Minimal Kiro footprint.
Enumerable work units	`docs/CODEBASE_AUDIT_ISSUES.md:1-243`	**Strong:** 12 numbered audit categories with file:line evidence and expected fixes — usable as an agent work queue. Module boundaries (`Discovery` , `Connect`) provide repeatable refactor targets.
Structural uniformity	`frontend/src/pages/ConnectPage.tsx` ↔ `DiscoveryPage.tsx` , Laravel controllers ↔ `dev-api/src/server.js`	**Weak for agents:** ~13% business-logic duplication across PHP and Node runtimes means every backend agent task must be applied twice or explicitly scoped to one runtime, doubling verification cost.
CI verifiability	`.github/workflows/ci.yml:1-58` , `backend/tests/Unit/HealthTest.php:9-14`	**Not ready:** PHPUnit tests assert hardcoded arrays without importing `App\` classes; no frontend tests; dev-api untested. Agents cannot rely on CI to catch regressions.

Honest assessment: The repo has better-than-average AI documentation (`AGENTS.md` + audit checklist) but is **not structurally ready** for unsupervised agent refactors until the dual runtime is consolidated and CI enforces lint + meaningful tests.

7.4 Database Usage

Check	Files Inspected	Finding	Consequence
Schema design	`docker/mariadb/init.sql:1-89`	Five tables in flat schema. FK constraints on `discovery_nodes.discovery_job_id` (`init.sql:38`) and `connect_check_results.connect_monitor_id` (`init.sql:66`). `users` table has no FK relationships. No secondary indexes on `status` , `discovery_job_id` parent lookups, or `checked_at` . MongoDB indexes in `docker/mongodb/init.js:47-48` and `dev-api/src/mongo.js:44-46` .	Referential integrity enforced only on two child tables; query performance on status/filter columns relies on full scans as data grows.
Migration hygiene	Tree scan — zero `backend/database/migrations/` files; `README.md:22-23` , `AGENTS.md:27`	Schema applied exclusively via manual `docker/mariadb/init.sql` . No versioned migrations, no rollback scripts, no `ALTER TABLE` guards. README explicitly documents "manual migration practice."	Any schema change requires editing init SQL and manual DBA coordination — agents cannot generate reversible migration files and CI cannot validate schema drift.
Data ownership	`docker/mariadb/init.sql:14-67` , architecture prior report	All five tables in one MariaDB database shared by Dashboard, Discovery, Connect, and Legacy modules. `users` table unused by application code.	80% shared-table coupling blocks safe per-domain extraction; agents refactoring Discovery can break Connect KPI queries in `DashboardController` .
Seed / sample data hygiene	`docker/mariadb/init.sql:71-89` , `docker/mongodb/init.js:3-45` , `dev-api/src/mongo.js:134-160`	MariaDB seeds use bare `INSERT` — re-running init on existing volume fails with duplicate key errors. MongoDB Docker init uses `insertMany` without idempotency guards. dev-api `seedMongoData()` in `dev-api/src/mongo.js:134-160` **is idempotent** (skips when documents exist; `--force` flag for re-seed). Sample email `admin@klearcom.local` — not production data.	Docker volume resets work; re-seeding MariaDB requires volume wipe. Inconsistent seed behavior between SQL and Node paths confuses onboarding.

7.5 Development Environment

Check	Files Inspected	Finding	Consequence	Next Step
`.env.example`	`backend/.env.example:1-16` , `dev-api/.env.example:1-2` , tree scan	Backend and dev-api examples present and referenced in `README.md:39-44` . **No `frontend/.env.example`**	Frontend contributors must read Docker Compose or README to discover	Add `front` `VITE_API_`

		despite `docker-compose.yml:73` setting `VITE_API_URL`.	`VITE_API_URL`; agent onboarding lacks a copy-paste template.	
OS portability	`README.md:28-36`, `package.json:5-7`	Quick-start path uses npm + Node only (no Docker required). `npm run dev` uses `concurrently` — cross-platform. Docker path optional via `docker compose up --build`.	Developers on Linux/macOS/Windows can run without vendor-specific tooling. No Windows blockers observed.	No action re onboarding
Containerization	`docker-compose.yml:1-77`, `docker/php/Dockerfile:1-18`, `frontend/Dockerfile:1-8`	Full stack: nginx, PHP-FPM 8.3, MariaDB 11, MongoDB 7, frontend Vite dev server. Health check on MariaDB (`docker-compose.yml:44-48`). DB ports exposed to host (`3306`, `27017`) — dev convenience, security debt.	Reproducible full-stack environment exists; port exposure is acceptable for local dev but should not ship to production compose.	Remove hos production o
Code style enforcement	Tree scan, `frontend/package.json:10-11`, `backend/composer.json:14-16`, `.github/workflows/ci.yml:47-58`	**Configured but largely unenforced:** No ESLint, Prettier, `.editorconfig`, or pre-commit hooks. PHPStan declared but unwired. Frontend `npm run build` runs `tsc --noEmit` (`frontend/package.json:11`) and CI executes build — **TypeScript type-checking is enforced**. No PHP lint in CI.	PHP and JS style drift immediately; `extract()` and class components persist despite AGENTS.md bans because nothing enforces them at commit/CI time.	Add `eslin` wire both int optional `.p`

No files matched `frontend/src/**/*.{tsx,jsx}` containing `style=\{\{`.

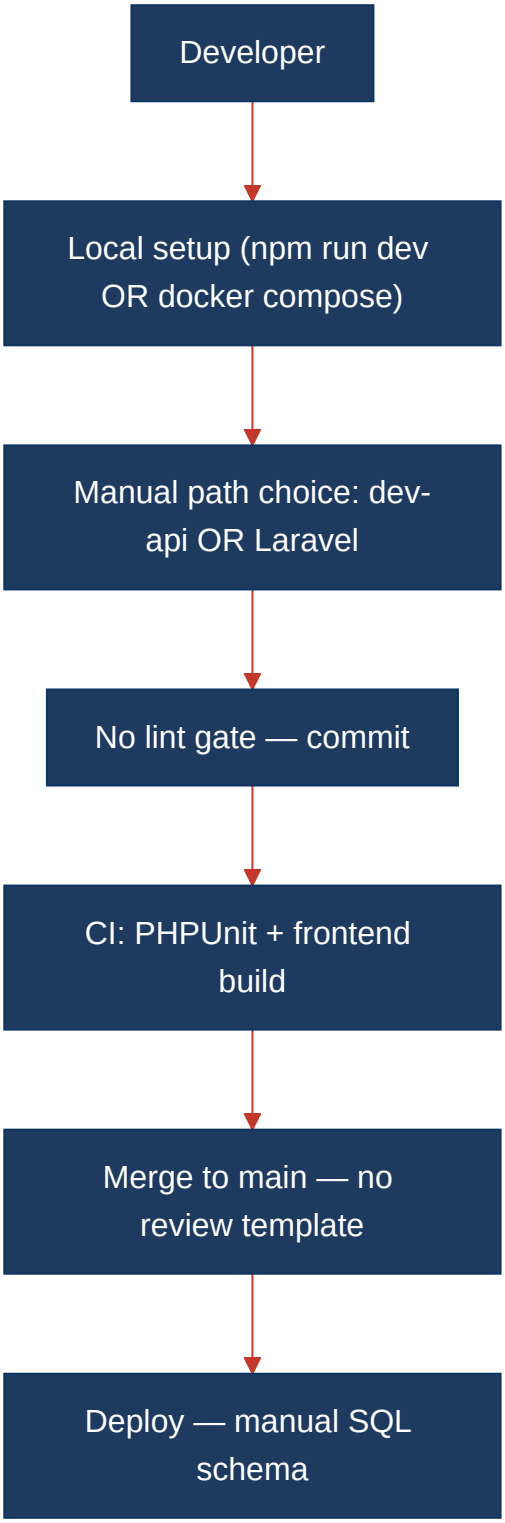
7.6 Prerequisites for Agentic Harness & Marketplace Readiness

Prerequisite	Current State	Gap
Enumerable work queue (clear, listable units of refactor work)	`docs/CODEBASE_AUDIT_ISSUES.md` lists 12 categories with file:line evidence; module `AGENTS.md` files scope Discovery/Connect	**MEDIUM** — dual runtime doubles every backend item; no machine-readable `tasks.json`
Isolated, verifiable units of work	Frontend modules cleanly separated; backend has 25 controller→model access points and 17 duplicated Laravel/dev-api capabilities	**HIGH** — shared DB + parallel API prevent isolated agent tasks
CI gate to accept agent-authored output	`.github/workflows/ci.yml` runs PHPUnit + `npm run build` (includes `tsc --noEmit`)	**CRITICAL** — no lint, no dev-api tests, 0% effective PHPUnit coverage of application code
Repo hygiene for automation (clean checkout, no secrets)	`.gitignore` covers env files; no committed secrets observed; lock files for npm committed	**MEDIUM** — missing `composer.lock` breaks PHP reproducibility

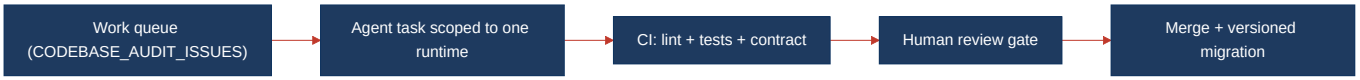
Marketplace packaging readiness	No OpenAPI spec, no auth, no package manifest, dual runtime	HIGH — cannot publish a single verifiable API contract

7.7 Diagrams

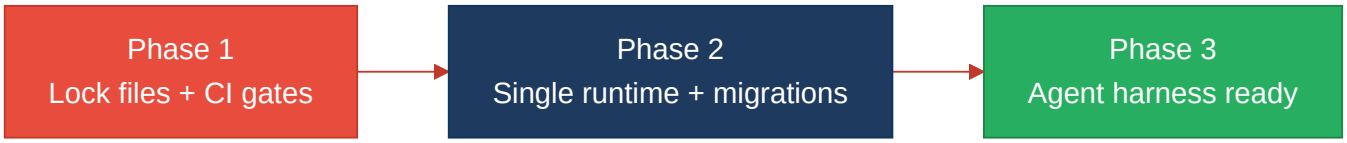
Current dev / delivery flow



Agentic harness readiness target



Improvement roadmap



7.8 Actions Required

Gap	Action	Rating	Priority
Missing `backend/composer.lock`	Run `composer update` in `backend/` and commit `backend/composer.lock`; add CI step verifying lock file is in sync with `composer.json`.	HIGH RISK	CRITICAL
Incomplete CI coverage	Extend `.github/workflows/ci.yml` with dev-api smoke test job, `vendor/bin/phpstan analyse`, ESLint on frontend, and `npm audit --audit-level=high`.	HIGH RISK	CRITICAL
No branch protection signals	Add `.github/CODEOWNERS` and `.github/pull_request_template.md`; enable required status checks on `main`.	HIGH RISK	HIGH
PHPStan declared but unwired	Create `backend/phpstan.neon` with level 5; add `vendor/bin/phpstan analyse` to CI; ban `extract()` via custom rules.	MODERATE	HIGH
Manual SQL schema — no migrations	Generate Laravel migrations from `docker/mariadb/init.sql`; restrict init.sql to first-boot seed; add migration step to CI backend job.	HIGH RISK	CRITICAL
Non-idempotent MariaDB seeds	Replace bare `INSERT` in `docker/mariadb/init.sql:71-89` with idempotent upserts or guard with `INSERT IGNORE`.	HIGH RISK	MEDIUM
Shared flat database schema	Assign table ownership per domain; plan schema split documented in init.sql header and module AGENTS.md files.	HIGH RISK	HIGH
Missing `frontend/.env.example`	Create `frontend/.env.example` with `VITE_API_URL=http://localhost:8080/api`; reference in README quick-start.	MODERATE	LOW
No ESLint / pre-commit enforcement	Add `eslint.config.js` with React/TS rules; add `.pre-commit-config.yaml` or enforce via CI only.	MODERATE	MEDIUM
Dual runtime blocks agent isolation	Deprecate dev-api or proxy to Laravel; until then, tag every audit item in `CODEBASE_AUDIT_ISSUES.md` with runtime scope.	MODERATE	CRITICAL
No observability baseline	Add Laravel `Log` facade usage in controllers/services; replace dev-api `console.log` with structured JSON logger; expose	HIGH RISK	MEDIUM

	`/api/health` metrics.		
No effective test gate for agents	Rewrite PHPUnit tests to import `App\` classes; add Vitest for frontend hooks; add dev-api route tests; target 80% coverage gate in CI.	HIGH RISK	CRITICAL

7.9 Expected Outcomes

- **Reproducible PHP installs:** Committing `backend/composer.lock` and CI lock-file verification eliminate floating dependency versions between contributors, CI, and agent-generated patches.
- **Trustworthy CI gate:** Lint (PHPStan + ESLint), dev-api smoke tests, and meaningful PHPUnit/Vitest coverage let an agentic harness accept auto-generated PRs with machine-verified quality signals.
- **Versioned schema evolution:** Laravel migrations replace manual init.sql edits, enabling agents to generate reversible schema changes with CI validation.
- **Single-runtime clarity:** Consolidating Laravel + dev-api removes duplicate verification and makes `CODEBASE_AUDIT_ISSUES.md` items independently actionable by agents.
- **Foundation for marketplace packaging:** OpenAPI spec + auth + enforced CI prerequisites unlock packaging Discovery and Connect modules as independently verifiable marketplace components.